Advances in SAM-Based Medical Image Segmentation: A Survey of Agentic, Weak-Supervision, and Preference Optimization Approaches

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Abstract—Medical image segmentation has undergone a paradigm shift driven by the emergence of foundation models, particularly the Segment Anything Model (SAM). This survey reviews three complementary research directions that collectively advance SAM-based segmentation beyond traditional supervised pipelines. First, *MedSAM-Agent* [1] reformulates segmentation as a multi-turn autonomous decision-making process, training a Multi-modal Large Language Model (MLLM) via Reinforcement Learning with Verifiable Rewards (RLVR) and a clinical-fidelity process reward to iteratively invoke SAM tools. Second, *All-in-SAM* [2] proposes a label-efficient pipeline in which SAM itself generates pseudo pixel-wise annotations from weak bounding-box supervision, which then serve as fine-tuning targets. Third, *MAPO* [3] introduces a Model-Agnostic Preference Optimization framework that leverages dropout-induced stochastic hypotheses to construct preference-consistent training signals applicable to any segmentation architecture. Experiments across 21 datasets and six imaging modalities demonstrate that these approaches consistently surpass state-of-the-art baselines. This review synthesizes their methodologies, benchmarks, and comparative findings, identifies key limitations, and outlines future research directions.

Index Terms—Medical image segmentation; Segment Anything Model (SAM); Agentic reinforcement learning; Multi-turn interaction; Weak annotation; Preference optimization; Multi-modal large language models; Foundation models.

I. INTRODUCTION

Medical image segmentation remains one of the most critical tasks in clinical computer vision, providing the quantitative delineation of anatomical structures and pathological regions required for diagnosis, treatment planning, and surgical guidance. Conventional approaches based on convolutional encoder-decoder architectures—most notably U-Net [4] and nn-UNet [5]—demonstrate strong performance on specific benchmarks; however, they require dedicated retraining for each task and modality, severely limiting clinical deployment breadth.

The introduction of the Segment Anything Model (SAM) [6] represented a landmark advance in generalizable segmentation. Trained on over one billion mask annotations, SAM achieves high-quality zero-shot segmentation conditioned on sparse human prompts such as points, bounding boxes, or coarse masks. Its successor SAM 2 [7] further extended these capabilities to video sequences. Despite these advances, direct application of SAM to medical images yields suboptimal performance due to the inherent domain gap [8]: medical imagery exhibits modality-specific noise, low contrast between tissues, and complex three-dimensional structures unseen during natural-image pre-training.

Three distinct research threads have emerged to address these limitations:

1. **Agentic reinforcement learning:** MedSAM-Agent [1] trains an MLLM to function as an autonomous agent that iteratively calls SAM tools, guided by a multi-dimensional clinical-fidelity reward.
2. **Weak-annotation exploitation:** All-in-SAM [2] exploits SAM's zero-shot edge-localization to bootstrap pixel-accurate pseudo labels from bounding-box annotations, then fine-tunes SAM on those labels in a label-efficient loop.
3. **Preference-based optimization:** MAPO [3] introduces a model-agnostic Direct Preference Optimization (DPO) framework that harnesses dropout-induced prediction diversity to form preference pairs, guiding any segmentation model toward structurally coherent outputs.

This survey provides a unified analysis of the three works. Section 2 surveys related literature. Section 3 details each methodology. Section 4 presents quantitative results and a comparison study. Section 5 discusses findings, limitations, future directions, and concludes.

II. LITERATURE REVIEW

A. Traditional Medical Image Segmentation

Deep learning-based medical image segmentation traces back to U-Net [4], whose skip-connected encoder-decoder architecture proved highly effective for biomedical images. Extensions such as Attention U-Net [9], TransUNet [10], Swin-UNet [11], UNETR [12], and SwinUNETR [13] progressively incorporated attention mechanisms and transformer backbones to better model long-range anatomical dependencies. nn-UNet [5] provides a self-configuring framework that automatically adapts architecture, preprocessing, and training to any segmentation dataset, setting strong baselines across medical benchmarks. Despite their performance, these models are inherently task-specific and require complete pixel-level annotation, motivating the search for generalizable and annotation-efficient alternatives.

B. The Segment Anything Model Family

SAM [6] established a prompting paradigm for zero-shot segmentation of natural images. Its architecture consists of a Vision Transformer (ViT) image encoder, a prompt encoder for

points, boxes, and masks, and a lightweight mask decoder. SAM 2 [7] extended the framework to video via a streaming memory mechanism. Domain-specific adaptations include MedSAM [8], SAM-Med3D [14], Medical SAM Adapter [15], and MedSAM2 [16]. Lightweight adaptation strategies such as SAM-Adapter [17] explore parameter-efficient domain alignment without full model retraining.

C. MLLMs for Segmentation

MLLMs have been integrated into segmentation workflows through two paradigms. Token-based methods such as LISA [18] introduce a special <seg> token whose embedding drives a pixel decoder. Grounding models like GLAMM [19] and HyperSeg [20] combine visual grounding with segmentation. In the medical domain, MedPLIB [21] and UniBiomed [22] extend these ideas with domain-specific supervision. A newer trend, exemplified by MedSAM-Agent [1] and SegAgent [23], treats SAM as an external tool invoked by the MLLM, preserving pixel-level precision while benefiting from MLLM reasoning.

D. Reinforcement Learning with Verifiable Rewards

Reinforcement learning from verifiable rewards (RLVR) emerged as a powerful post-training paradigm for LLMs following DeepSeekMath [24]. In the visual domain, RLVR has been applied to reasoning segmentation (Seg-R1 [25], Seg-Zero [26]), grounding [27], and medical reasoning (MedVLM-R1 [28], Med-RLVR [29]). MedSAM-Agent [1] extends this paradigm to multi-turn tool use with process-level supervision, going beyond single-pass outcome rewards.

E. Weak Supervision and Annotation Efficiency

Annotation cost is a perennial bottleneck in medical AI. Semi-supervised methods such as Mean Teacher [30] and UA-MT [31] leverage consistency regularization. Self-supervised approaches [32] utilize unlabeled data through contrastive pretext tasks. A key finding in prior work [33] was that SAM achieves near-pixel accuracy given bounding-box prompts, making it a natural pseudo-labeler. All-in-SAM [2] formalizes this into a full annotation-and-finetuning pipeline.

F. Preference Optimization in Vision

Direct Preference Optimization (DPO) [34] and related methods (IPO [35], KTO [36], ORPO [37]) emerged as efficient alternatives to RLHF for aligning LLMs with human preferences. Their extension to visual segmentation is nascent: SAMPO [38] applies preference optimization to SAM for intent-aware segmentation, and Konwer et al. [39] combine unsupervised prompts with DPO-based alignment for semi-supervised medical segmentation. MAPO [3] generalizes these ideas to any segmentation architecture via dropout-driven hypothesis generation.

III. METHODOLOGY

A. MedSAM-Agent: Agentic Multi-turn Reinforcement Learning

Problem Formulation: MedSAM-Agent [1] reformulates medical image segmentation as a Markov Decision Process (MDP).

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nforcement Learning

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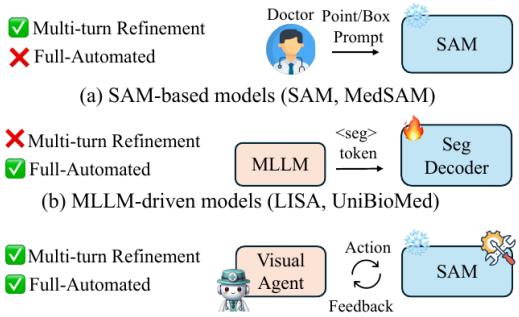


Figure 1. Comparison of segmentation paradigms [1]. (a) SAM-based models require continuous manual prompting. (b) MLLM-driven models use <seg> tokens and pixel decoders. (c) MedSAM-Agent acts as an autonomous visual agent performing multi-turn refinement through iterative feedback and tool interaction.

Given an image $\mathbf{I}$ and a textual target prompt P, a policy model π_θ (Qwen3-VL-8B) produces action a_t at each time step t. Actions belong to one of three types: a bounding-box prompt $a_t{=}[x_1, y_1, x_2, y_2]$, a point click $a_t{=}[x_1, y_1]$ with polarity $\alpha{\in}\{+1, -1\}$, or a stop signal. The observation o_t is the updated mask $M_t = F_{\text{seg}}(\mathbf{I}, o_{t-1}, a_t)$ produced by the interactive backend. The state aggregates all prior action-observation pairs:

$$s_t = \{(a_0, o_0), (a_1, o_1), \ldots, (a_{t-1}, o_{t-1})\}. \tag{1}$$

Expert Trajectory Generation: A hybrid prompting strategy generates diverse, high-quality trajectories. In the *Box-to-Point* workflow, a bounding box with random jitter $\varepsilon{\sim}\mathcal{U}(-\delta, \delta)$ initializes the trajectory; corrective clicks are placed at the centroid of the largest False Negative (FN) or False Positive (FP) cluster via distance transform. A *Sequential-Click* paradigm starts from the target centroid instead. A progress-constrained mechanism enforces a minimum IoU gain per action:

$$\Delta\text{IoU}_t = \text{IoU}(M_t, M_{\text{target}}) - \text{IoU}(M_{t-1}, M_{\text{target}}) \geq \tau, \quad \tau{=}0.04. \tag{2}$$

If the candidate fails this threshold the simulator resamples up to $N{=}5$ times, producing a dataset $\mathcal{D}_{\text{traj}}$ of 449K monotonically improving sequences.

Two-Stage Training Pipeline: Stage 1 – SFT Cold-Start. Expert trajectories are formatted as structured tool-call sequences wrapped in <tool_call> tags. The SFT loss minimizes the

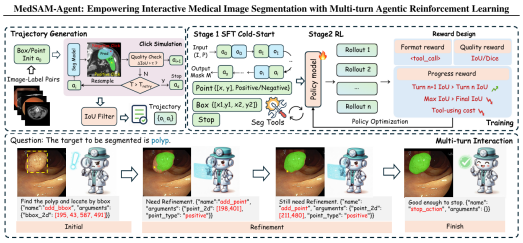


Figure 2. **Overview of MedSAM-Agent.** We develop a hybrid prompting strategy for expert-curated trajectory generation that transforms image-label pairs into high-quality interaction sequences via simulated clicks and IoU-based filtering. Then these trajectories support

Figure 2. Overview of MedSAM-Agent [1]. *Left:* Hybrid prompting strategy transforms image-label pairs into high-quality interaction trajectories via simulated clicks and IoU-based filtering. *Center:* Two-stage training pipeline (Stage 1 SFT cold-start; Stage 2 RL). *Right:* Fine-grained reward design.

negative log-likelihood over the full trajectory:

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\mathbb{E}_{(\mathbf{I},P,y)\sim\mathcal{D}_{\mathrm{sft}}}\left[\sum_{t=1}^{|y|}\log \pi_\theta(y_t \mid \mathbf{I}, P, y_{<t})\right]. \quad (3)$$

Stage 2 – Reinforcement Learning. Group Relative Policy Optimization (GRPO) [24] is applied with a composite terminal reward:

$$R_{\mathrm{total}} = w_1 R_{\mathrm{fmt}} + w_2\,\mathrm{clip}(R_{\mathrm{qual}} + \lambda_1 R_{\mathrm{imp}} - \lambda_2 R_{\mathrm{over}} - \lambda_3 R_{\mathrm{cost}}, 0, 1), \quad (4)$$

where R_{fmt} enforces format adherence; $R_{\mathrm{qual}}{=}w_{\mathrm{iou}}\cdot \mathrm{IoU}_{\mathrm{final}} + w_{\mathrm{dice}}\cdot \mathrm{Dice}_{\mathrm{final}}$ measures segmentation quality; $R_{\mathrm{imp}}{=}\sum_t \max(0, \mathrm{IoU}_t{-}\mathrm{IoU}_{t-1})$ rewards monotonic progress; $R_{\mathrm{over}}{=}\mathrm{IoU}_{\max}{-}\mathrm{IoU}_{\mathrm{final}}$ penalizes quality degradation after peak; and R_{cost} penalizes trajectory length. Within each group of G rollouts, the normalized advantage is:

$$A_i = \frac{R_{\mathrm{total},i} - \mathrm{mean}(R_{\mathrm{total}})}{\mathrm{std}(R_{\mathrm{total}})}. \quad (5)$$

B. All-in-SAM: Weak Annotation to Pixel-wise Segmentation

SAM-Empowered Annotation (Stage 1): All-in-SAM [2] uses the pretrained SAM as both annotation engine and segmentation backbone. Bounding boxes derived from weak annotations are provided as sparse prompts to the *frozen* SAM, which generates pseudo pixel-level masks $\hat{M}$. This exploits SAM's strong edge-detection capability: prior work [33] reports a Dice of 0.883 on MoNuSeg when bounding-box prompts are supplied—far exceeding supervised baselines without any fine-tuning.

Prompt-Based Fine-Tuning (Stage 2): SAM is fine-tuned using the SAM-Adapter strategy [17]: high-frequency texture features extracted from each image are injected additively into multiple encoder layers, and the prompt encoder together with the lightweight decoder become learnable. Formally, the adapted embedding at layer l is:

$$\tilde{z}^{(l)} = z^{(l)} + f_\phi(h_{\mathrm{HF}}(\mathbf{I}))\,, \quad (6)$$

where $z^{(l)}$ is the original ViT embedding, h_{HF} extracts high-frequency features, and f_ϕ is a lightweight convolutional projection. The fine-tuned model operates *without any prompt at inference time*, having internalized the spatial prior of the target domain.

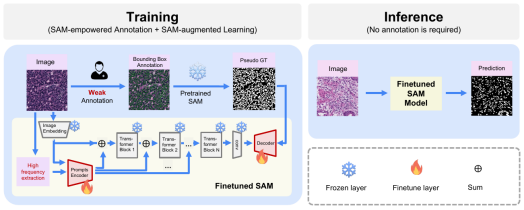


Fig. 2. The proposed pipeline using weak and few annotations for nuclei segmentation. In the training stage, only the bounding boxes (in green color) of nuclei were provided as the weak annotation label to generate the approximate segmentation masks. Then, with the supervision of the approximate segmentation masks, the prompt-based finetuning was applied to the pretrained SAM model. In the inference stage, nuclei can be segmented directly from images without box prompts.

may exhibit suboptimal performance or even fail. Nonetheless, two subsets, with 24 images assigned to the training set and

Figure 3. All-in-SAM training and inference pipeline [2]. Bounding boxes (green) serve as weak annotations; the frozen pretrained SAM generates pseudo ground-truth masks; the fine-tuned SAM with prompt adapters performs inference without manual prompts. Snowflake: frozen layers; flame: trainable layers.

C. MAPO: Model-Agnostic Preference Optimization

Dropout-Driven Preference Data Generation: MAPO [3] applies dropout at inference with rates $\mathcal{D}{=}\{d_1,\ldots,d_M\}$, generating K diverse prediction candidates $\{\hat{y}_k\}_{k=1}^K$ per training image. The preferred (good) candidate is:

$$k^+ = \arg\max_k \mathrm{Dice}(\hat{y}_k, y^{\mathrm{gt}}), \quad \hat{y}^+ = \hat{y}_{k^+}. \quad (7)$$

A valid candidate set enforces a minimum Dice gap τ from $\hat{y}^+$:

$$\mathcal{C} = \left\{k \mid \mathrm{Dice}(\hat{y}^+, y^{\mathrm{gt}}) - \mathrm{Dice}(\hat{y}_k, y^{\mathrm{gt}}) \geq \tau\right\}. \quad (8)$$

The dispreferred (bad) candidate is $k^-{=}\arg\min_{k\in\mathcal{C}} \mathrm{Dice}(\hat{y}_k, y^{\mathrm{gt}})$.

Preference Optimization Objective: The voxel-wise Bernoulli log-likelihood of prediction $\hat{y}$ is:

$$\log \pi_\theta(\hat{y} \mid \mathbf{x}) = \frac{1}{|\Omega|}\sum_{i\in\Omega}\left[\hat{y}_i \log p_{\theta,i} + (1{-}\hat{y}_i)\log(1{-}p_{\theta,i})\right], \quad (9)$$

where Ω is the set of spatial indices and $p_\theta{\in}[0,1]^{|\Omega|}$. The combined training objective is:

$$\mathcal{L} = \lambda\left(\mathcal{L}_{\mathrm{Dice}}(p_\theta, y^{\mathrm{gt}}) + \mathcal{L}_{\mathrm{CE}}(p_\theta, y^{\mathrm{gt}})\right) + \mathcal{L}_{\mathrm{DPO}}, \quad (10)$$

where $\mathcal{L}_{\mathrm{DPO}}$ is the standard DPO log-ratio loss [34] with inverse temperature $\beta{=}0.1$ and $\lambda{=}0.5$.

Online Preference Training: Preference pairs are regenerated every 50 epochs (2D) or 200 epochs (3D) as the model improves, ensuring supervision remains aligned with the model's current capability without additional human intervention.

IV. RESULTS

A. MedSAM-Agent Quantitative Results

MedSAM-Agent [1] was evaluated across 21 open-source datasets spanning six modalities: CT, MRI, X-Ray, Ultrasound, Fundus, and Endoscopy. Table 1 reports mean Dice and IoU

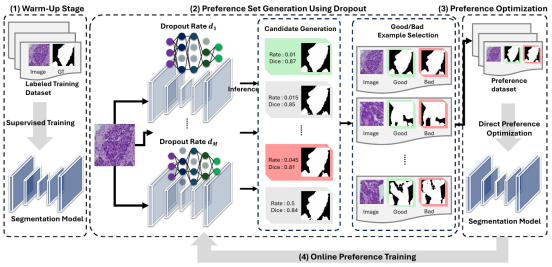


Figure 1. Overview of our proposed model-agnostic preference optimization for medical image segmentation, which consists of 4 stages: (1) Warm-up stage, (2) Preference set generation using dropout, (3) preference optimization, (4) Online preference training.

Figure 4. Overview of MAPO [3]: (1) Warm-up supervised training; (2) Preference set generation via dropout-based stochastic inference; (3) Direct Preference Optimization; (4) Online iterative preference training. The framework is fully architecture-agnostic (2D CNN, ViT, hybrid, 3D).

for each modality and compares MedSAM-Agent against interactive baselines (SAM2, MedSAM, IMISNet) and MLLM-based automated methods. Ours-MedSAM2 achieves state-of-the-art among all fully automated methods, surpassing interactive baselines using ground-truth box prompts in several modalities.

The ablation study in Table 2 confirms the contribution of each component. The hybrid prompting strategy outperforms single-paradigm approaches, and the full SFT+RL pipeline achieves the best balance of segmentation quality and interaction efficiency (average 2.11 turns). The process rewards R_{imp}, R_{over}, and R_{cost} each contribute measurably, with R_{imp} having the largest impact (Dice: $0.794 \rightarrow 0.772$ upon removal).

B. *All-in-SAM Quantitative Results*

All-in-SAM [2] was evaluated on the MICCAI 2018 MoNuSeg nuclei segmentation benchmark. Table 3 shows that the proposed pipeline achieves 82.54% Dice with full complete annotation, outperforming nn-UNet (82.44%) and BEDs (81.77%). With only 4% of training data and *weak* (bounding-box only) annotation, it achieves 80.99% Dice— comparable to nn-UNet trained on the full dataset with complete annotations (82.12%).

C. *MAPO Quantitative Results*

MAPO [3] was evaluated on five 2D and three 3D medical segmentation datasets using seven backbone architectures. Tables 4 and 5 show consistent Dice and Average Surface Distance (ASD) improvements across all architectures. The largest gains occur on architectures with unstable baseline training: TransAttUNet + MAPO achieves +28.93% Dice on the Kvasir dataset, demonstrating that preference optimization acts as a powerful training stabilizer.

D. *Comparison Study*

Table 6 provides a unified cross-method comparison of all three surveyed frameworks along eight key dimensions. The three works occupy largely orthogonal niches: MedSAM-Agent addresses the *autonomy* of interaction, All-in-SAM addresses *annotation efficiency*, and MAPO addresses *training alignment and architectural generalization*. A natural synthesis

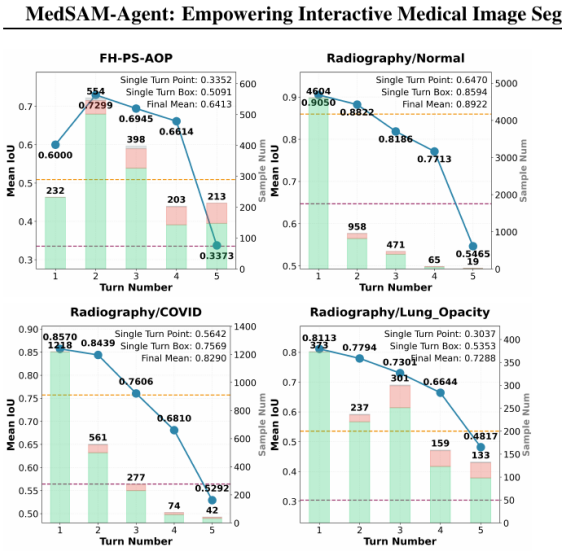


Figure 3. **Analysis of multi-turn interaction.** The orange and purple lines represent the performance of the static Single-Turn Box and Single-Turn Point prompts, respectively, where inputs are derived from ground-truth bounding boxes and centroids. The blue line plots the Mean IoU across successive interaction turns. The bar charts illustrate the distribution of sample outcomes at each turn, where green, red, and grey segments denote the proportion

Figure 5. Multi-turn interaction analysis of MedSAM-Agent [1] on FH-PS-AOP and Radiography datasets. Blue line: mean IoU per interaction turn. Orange/purple dashed lines: single-turn Box/Point upper bounds. Green/red/grey bars: proportions of improving, declining, or unchanged IoU samples per turn. The agent consistently surpasses the static single-turn upper bound.

Table 1. Quantitative Comparison Across Six Medical Imaging Modalities (MedSAM-Agent [1]). Mean Dice / IoU per modality and average. Point and Box denote prompts from ground-truth centroids and bounding boxes, respectively. **Bold**: best non-interactive result; underline: second-best.

Method	CT		MRI		X-Ray		Ultrasound		Fundus		Endoscopy		Avg	
	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
SAM-based methods (interactive)														
SAM2-Point [7]	0.610	0.531	0.488	0.412	0.546	0.422	0.528	0.408	0.362	0.265	0.659	0.597	0.532	0.439
SAM2-Box [7]	0.837	0.754	0.838	0.748	0.827	0.733	0.871	0.779	0.717	0.574	0.922	0.872	0.835	0.743
MedSAM-Point [16]	0.747	0.640	0.667	0.584	0.843	0.776	0.591	0.467	0.414	0.346	0.822	0.736	0.681	0.592
MedSAM-Box [16]	0.847	0.759	0.854	0.768	0.897	0.836	0.890	0.806	0.846	0.743	0.925	0.872	0.876	0.797
MLLM-based methods (automated)														
LISA [18]	0.088	0.051	0.110	0.063	0.322	0.230	0.385	0.280	0.036	0.019	0.282	0.222	0.204	0.144
GLAMM [19]	0.088	0.053	0.106	0.061	0.285	0.194	0.402	0.294	0.027	0.014	0.245	0.179	0.192	0.132
HyperSeg [20]	0.098	0.069	0.153	0.101	0.336	0.231	0.434	0.336	0.023	0.012	0.396	0.342	0.240	0.182
Seg-R1 [25]	0.099	0.064	0.126	0.075	0.417	0.318	0.490	0.382	0.455	0.325	0.533	0.471	0.353	0.272
MedPLIB [21]	0.052	0.038	0.177	0.136	0.115	0.075	0.088	0.056	0.298	0.201	0.132	0.097	0.144	0.101
UniBiomed [22]	0.724	0.634	0.807	0.716	0.817	0.748	0.736	0.618	0.794	0.668	0.778	0.704	0.776	0.681
Citrus-V	0.363	0.302	0.326	0.258	0.624	0.536	0.134	0.094	0.782	0.655	0.666	0.616	0.482	0.410
MedSAM-Agent (proposed)														
Ours-IMISNet	0.732	0.654	0.783	0.685	0.798	0.723	0.658	0.547	0.803	0.678	0.803	0.736	0.763	0.670
Ours-MedSAM2	**0.720**	**0.633**	**0.793**	**0.701**	**0.833**	**0.775**	**0.793**	**0.685**	**0.813**	**0.692**	**0.811**	**0.744**	**0.794**	**0.705**

Table 2. Ablation Study of MedSAM-Agent [1] (Segmentation tool: MedSAM2).

Setting	Dice	IoU	n_{turn}
Qwen3-VL-8B base	0.263	0.178	4.34
+ Cold-start SFT only (Hybrid)	0.769	0.686	2.35
– only Sequential-Click	0.719	0.623	2.94
– only Box-to-Point	0.745	0.649	2.06
+ RL only	0.678	0.593	2.23
+ SFT + RL (Full)	**0.794**	**0.705**	**2.11**
– w/o R_{imp}	0.772	0.688	2.08
– w/o R_{over}	0.785	0.693	2.24
– w/o R_{cost}	0.788	0.696	2.26

would combine autonomous multi-turn prompting with weak-annotation bootstrapping and preference-guided optimization into a single unified training pipeline.

Table 3. Nuclei Segmentation on MoNuSeg [2]. **Red/bold**: best; blue: second-best.

Label	Method	Data	Dice(%)	IoU(%)
Complete	Xie et al. [32]	All	—	—
	LViT [40]	All	80.33	67.24
	BEDs [41]	All+	81.77	—
	nnUNet [5]	All	82.44	**69.76**
	nnUNet [5]	4%	79.20	65.40
	Proposed [2]	All	**82.54**	69.74
	Proposed [2]	4%	81.34	68.10
Weak	nnUNet [5]	All	82.12	69.35
	nnUNet [5]	4%	79.13	65.27
	Proposed [2]	All	**82.46**	69.73
	Proposed [2]	4%	80.99	67.60

Table 4. Dice (%) of MAPO [3] on Five 2D Medical Image Segmentation Datasets.

	Method	ISIC	EBHI	FIVES	Kvasir	COVID-QU-Ex
CNN	U-Net	81.68	91.84	86.58	65.06	93.65
	U-Net+MAPO	82.80	91.96	88.12	74.08	94.99
ViT	SwinUNet	78.65	90.75	82.79	52.20	92.61
	SwinUNet+MAPO	81.98	92.25	85.48	59.83	94.67
Hybrid	TransAttUNet	81.65	91.17	75.19	53.80	95.13
	TransAttUNet+MAPO	**84.94**	**92.95**	75.99	**81.73**	**96.30**

Table 5. Dice (%) and ASD of MAPO [3] on Three 3D Medical Image Segmentation Datasets.

	Method	Parse2022		Sliver07		Luna16	
		Dice	ASD	Dice	ASD	Dice	ASD
CNN	VNet	66.25	13.20	71.23	30.79	96.98	2.18
	VNet+MAPO	68.79	12.90	73.40	27.94	96.99	1.56
	DynUNet	72.95	17.78	62.90	27.69	95.96	5.10
	DynUNet+MAPO	74.20	15.20	65.25	21.22	94.65	4.90
ViT	DFormer	65.57	15.71	67.60	26.87	96.98	3.93
	DFormer+MAPO	67.60	13.11	69.50	21.11	**97.02**	**3.90**
Hyb.	UNETR	60.01	18.15	72.31	26.27	96.47	1.32
	UNETR+MAPO	**62.76**	**15.47**	**74.93**	**21.35**	97.01	4.26

Table 6. Cross-Method Comparison of MedSAM-Agent [1], All-in-SAM [2], and MAPO [3] Across Eight Key Dimensions.

Dimension	**MedSAM-Agent** [1]	**All-in-SAM** [2]	**MAPO** [3]
Core Idea	MLLM autonomous agent invoking SAM iteratively via RL	SAM generates pseudo pixel-labels from bounding boxes then fine-tunes itself	Dropout-driven preference pairs align any segmentation model via DPO
Training Paradigm	SFT cold-start + RLVR (GRPO) with process reward	SAM-Adapter fine-tuning on pseudo labels	Warm-up supervised + iterative online DPO
Annotation Required	Text prompt only; trajectories synthetically generated	Bounding boxes only (weak)	Full pixel labels for warm-up; preferences auto-generated
Interaction Type	Multi-turn autonomous (box + point + stop)	Single-pass; no inference prompt required	Single-pass inference; preference-based training
Architecture Scope	Requires SAM-family backend; MLLM policy is modular	Fine-tunes SAM with lightweight adapters	Fully agnostic: 2D CNN, ViT, 3D models
Best Performance	Avg Dice 0.794, IoU 0.705 (6 modalities, 21 datasets)	Dice 82.54% (MoNuSeg full); 80.99% (4%, weak labels)	Dice gains +0.1%–+28.9% across 8 datasets, 7 architectures
Key Strength	Surpasses static interactive upper bounds; tool-agnostic at inference	Label-efficient; competitive with SOTA at 0.5% data	Universal backbone coverage; stabilizes unstable training
Key Limitation	Multi-turn inference latency; 2D-only	SAM-dependent; validated on one 2D benchmark	Requires pixel labels for warm-up; online regeneration overhead
Datasets / Modalities	21 datasets: CT, MRI, X-Ray, US, Fundus, Endoscopy	MoNuSeg (histopathology)	5×2D + 3×3D (dermoscopy, histology, fundus, colonoscopy, CT, MRI)

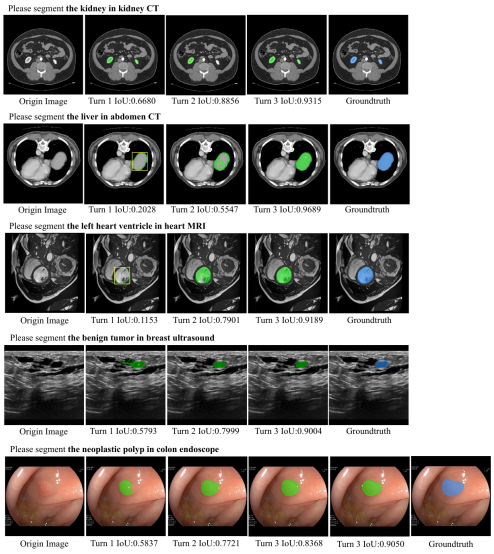


Figure 4. **Case study.** Yellow boxes indicate bounding box prompts, yellow points represent positive clicks, and red points denote negative clicks.

Figure 6. Qualitative case study of MedSAM-Agent [1] (kidney CT, liver CT, heart MRI, breast ultrasound, colon endoscopy). Columns show the original image and successive interaction turns ending at ground truth. Yellow boxes: bounding-box prompts; yellow points: positive clicks; red points: negative clicks. IoU improves monotonically across all modalities.

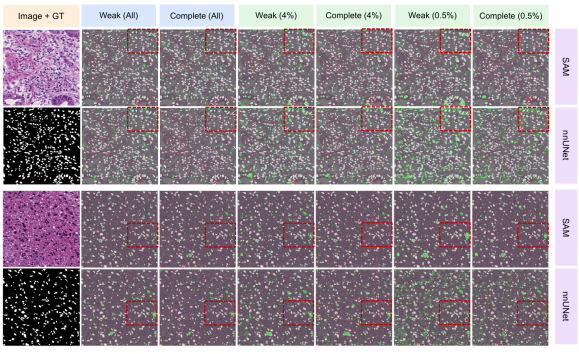


Fig. 3. Qualitative results of nuclei segmentation of different models trained by different numbers of annotated data. Upper row: proposed method; lower row: nnUNet. The mask region in green color indicates the false positive of prediction, the white color indicates the true positive, and the red color indicates the false negative.

Figure 7. Qualitative nuclei segmentation results [2]. *Top rows:* proposed method; *bottom rows:* nnUNet [5]. Columns correspond to image+GT and results at 100%, 4%, and 0.5% data with weak vs. complete annotation. Green: false positive; white: true positive; red: false negative.

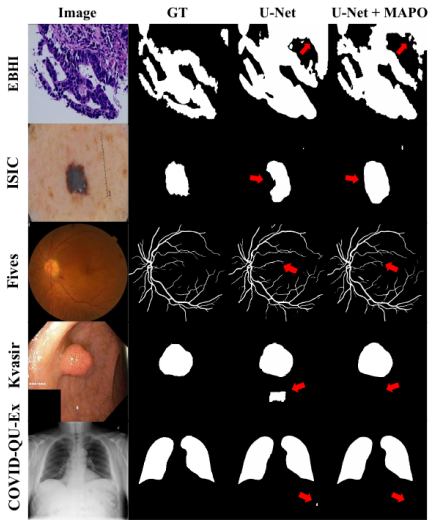


Figure 4. Qualitative comparison between our proposed method and baseline using U-Net [49] architecture.

Loss function	RollingUNet [38]	SwinUNet [5]	TransattUNet[7]
CE[40]	86.97	90.27	88.40
Focal[37]	86.45	78.39	88.27
DiceCE[16]	87.01	90.75	91.17
DiceFocal[37]	86.81	91.11	91.10
Dice[42]	87.79	78.39	89.38
Lovasz[4]	86.74	91.02	88.79
HD [6, 30]	83.17	83.11	82.18
MAPO (Ours)	90.32	92.25	92.95

Table 5. Dice similarity scores of RollingUNet [38], SwinUNet [5], and TransattUNet[7] using various loss functions on EBHI

Figure 8. Qualitative comparison: U-Net vs. U-Net+MAPO [3] on five 2D datasets. MAPO consistently produces fewer false positives and more refined contour boundaries (red arrows mark typical improvements).

V. DISCUSSION AND CONCLUSION

A. Key Findings

Multi-turn interaction outperforms single-turn prompting. MedSAM-Agent [1] demonstrates that an MLLM-guided agent performing iterative box-and-point refinement consistently exceeds even oracle single-turn bounding-box prompts. The progressive improvement bonus R_{imp} in Eq. (4) is the most critical process-reward term: its removal causes a Dice drop from 0.794 to 0.772, confirming that rewarding each incremental step is more effective than terminal outcome supervision alone.

SAM's edge-localization enables label-efficient learning. All-in-SAM [2] demonstrates that SAM's zero-shot precision, conditioned on bounding boxes, generates pseudo masks rivaling full pixel-wise annotations in quality. Training with only 4% data and weak labels (80.99% Dice) nearly matches full-data supervised nn-UNet (82.44%). This has significant implications for clinical settings where annotation resources are scarce.

Preference optimization provides architecture-agnostic improvements. MAPO [3] shows that DPO-based alignment combined with dropout-induced diversity consistently improves both Dice and boundary precision (ASD) across CNN, Transformer, and 3D architectures. The largest gains occur on architectures with training instability (up to +28.93% Dice on Kvasir), revealing preference optimization as a powerful regularizer beyond an alignment mechanism.

B. Limitations

MedSAM-Agent is limited to 2D image slices. Multi-turn interaction for volumetric data introduces additional complexity in spatial memory and cross-slice consistency. Inference latency grows linearly with the number of interaction turns, which may hinder real-time clinical use.

All-in-SAM has been validated only on the MoNuSeg nuclei segmentation benchmark. Generalizability of pseudo-label quality to other modalities (CT/MRI volumes, irregular lesion boundaries) requires further investigation. The framework also inherits SAM's limitations in low-contrast or out-of-distribution scenarios.

MAPO requires a ground-truth-supervised warm-up stage, limiting its use in purely weakly supervised settings. Online preference regeneration introduces computational overhead, and the framework has not been evaluated on interactive or agentic segmentation settings.

C. Future Directions

1. **3D/volumetric extension:** Extending agentic segmentation to volumetric modalities requires a 3D state representation, cross-slice attention, and volumetric IoU rewards. MedSAM2 [16] provides a natural backend.

2. **Unified annotation-and-training loop:** Combining All-in-SAM's pseudo-labeling with MedSAM-Agent's agentic refinement could yield a fully autonomous pipeline starting from zero annotation and progressively improving both pseudo labels and policy.

3. **Preference optimization for interactive segmentation:** Applying MAPO's preference framework to multi-turn trajectories from MedSAM-Agent could provide richer supervision and reduce required interaction turns.

4. **Computational efficiency:** Early-exit mechanisms, speculative decoding, and KV-cache optimization are needed for near-real-time clinical deployment of multi-turn agents.

5. **Large-scale weak-label pretraining:** Scaling All-in-SAM's pipeline to millions of images across diverse modalities could yield a universally generalizable medical SAM without pixel-level annotation.

D. Conclusion

This survey reviewed three complementary advances in SAM-based medical image segmentation. MedSAM-Agent [1] establishes agentic reinforcement learning as a viable paradigm for autonomous multi-turn segmentation, achieving state-of-the-art Dice of 0.794 across six imaging modalities. All-in-SAM [2] demonstrates that SAM's zero-shot capability can close the gap between weak and strong annotation, enabling competitive performance with as little as 0.5% of labeled data. MAPO [3] introduces a model-agnostic preference optimization framework that consistently improves segmentation quality and boundary precision across diverse architectures. Together, these works point toward a future of annotation-efficient, autonomously-refined, and preference-aligned medical AI systems capable of clinical deployment across the full spectrum of imaging modalities.

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